BIO 233 | Study Assistance Sheet | Chapters 12 + 13

ONE SIDE ONLY • Text may be printed. Any diagrams must be drawn by hand. • US Letter landscape; print at 100%.

ORGANIZATION + CELLS

SAME: Sensory = Afferent (arrives); Motor = Efferent (exits). CNS: brain + cord. PNS: peripheral nerves. ENS: gut-wall networks.

Somatic motor: skeletal muscle, including reflexes. **Autonomic:** smooth/cardiac muscle, glands, adipose; sympathetic + parasympathetic.

Receptors: extero = external; intero = internal; proprio = muscle/joint position.

Neuron = one cell; **nerve** = PNS axon bundle; **tract** = CNS axon bundle. **Ganglion** = PNS cell bodies; **nucleus** = CNS cell bodies.

Soma = body; perikaryon = cytoplasm around nucleus. Dendrites receive; axon carries; telodendria end in terminals. Axolemma = membrane; axoplasm = cytoplasm. Nissl = rough ER + ribosomes.

Kinesin: anterograde, away from soma. **Dynein:** retrograde, toward soma. Transport uses ATP.

Shapes: anaxonic, no obvious axon; bipolar, two processes; unipolar, soma off to side (sensory); multipolar, many dendrites + one axon (motor examples AND interneurons).

GLIA + MYELIN

CNS: Astrocytes support/BBB/local environment/scar; Ependymal line ventricles + canal, help with CSF; Oligodendrocytes myelinate; Microglia phagocytose.

PNS: Satellite cells surround ganglion cell bodies; Schwann cells surround axons and form myelin.

Oligo: segments on several axons. **Schwann:** one myelinated segment on one axon. Internode = myelinated segment; node = gap. White matter: many myelinated axons; gray: many cell bodies/processes.

Repair: distal Wallerian degeneration; macrophage cleanup; Schwann guide; axon regrowth. CNS repair limited by complexity, scar and inhibitory signals.

MEMBRANE POTENTIALS

Rest about -70 mV. Na+ high outside, K+ high inside. Negative interior relative to exterior. **Pump: 3 Na+ OUT, 2 K+ IN, 1 ATP.**

K+: chemical force out, electrical force in. **Na+:** both forces inward. Equilibrium examples: K+ -90 mV; Na+ +66 mV. These are selective-permeability examples, not a universal "pump off" result.

Channels: leak open at rest; chemical/ligand gated bind ligand; voltage gated respond to voltage; mechanical gated respond to distortion.

Depolarize: more positive. **Repolarize:** return toward rest. **Hyperpolarize:** more negative than rest. Depolarization alone does not prove an AP.

Graded: local, variable, decreases with distance; can depolarize or hyperpolarize; sums; no AP threshold or refractory period of its own.

AP: axon/initial segment; threshold, all-or-none, regenerated without fading, refractory period, no amplitude summation.

Sequence: -70 rest; about **-55 threshold** for lecture; Na+ activation/influx; about +30 peak; Na+ inactivation + K+ efflux; repolarization; delayed K+ closure gives undershoot.

Absolute refractory: cannot fire again. **Relative:** stronger stimulus needed. Recently active membrane favors forward propagation.

Continuous: adjacent unmyelinated regions. **Saltatory:** regenerate at nodes. Myelin + bigger diameter increase speed. A: large/myelinated, up to 268 mph; B: smaller/myelinated, about 40; C: unmyelinated, about 2.

Intensity: stronger sustained input usually increases AP frequency, not AP height. Absolute refractory limits maximum firing rate.

NEURAL CIRCUITS

Divergence: one to many. **Convergence:** many to one. **Serial:** sequence. **Parallel:** simultaneous. **Reverberation:** feedback loop.

SYNAPSES + SUMMATION

Electrical: gap junctions. **Chemical:** transmitter crosses cleft. Axodendritic = to dendrite; axosomatic = to soma; axoaxonic = to axon.

ACh sequence: AP arrives; voltage-gated Ca2+ opens; Ca2+ influx triggers vesicle exocytosis; ACh binds postsynaptic cation channels; net depolarization. AChE splits ACh into acetate + choline; choline is recycled.

Delay: Ca2+/release steps; more synapses add delay. **Fatigue:** demand exceeds transmitter replenishment.

Cholinergic: ACh. **Adrenergic:** NE/related signaling. Dopamine, serotonin, NE are amines; endorphins/enkephalins/dynorphins are opioid peptides. GABA/glycine inhibit; glutamate often excites. Receptor matters.

Ionotropic: direct channel effect. **Metabotropic:** receptor/G protein; adenylate cyclase converts ATP to cAMP (second messenger); downstream channels/enzymes change.

EPSP: graded excitation, closer to threshold. **IPSP:** inhibitory graded effect; chapter uses hyperpolarization. K+ out or appropriate Cl- channels can inhibit.

Temporal: repeated input at one site. **Spatial:** multiple sites together. Graded potentials sum; AP amplitudes do not. -70 + 15 = -55; -85 + 15 = -70.

Presynaptic inhibition: GABA example reduces Ca2+ entry and release. **Facilitation:** serotonin example increases Ca2+ entry and release.

CORD ANATOMY

31 pairs: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, 1 coccygeal. C1-7 above matching vertebra; C8 between C7/T1; T1 onward below. Conus ends cord near L1-2; cauda = long roots; filum = support thread.

Dura, arachnoid, pia (outer to inner). Epidural outside dura; CSF in subarachnoid + central canal. Denticulate ligaments: pia to dura, lateral stability; filum/coccygeal support: longitudinal stability.

ROOTS, RAMI + PLEXUSES

DAVE: dorsal afferent; ventral efferent. DRG has sensory cell bodies. Spinal nerves and anterior/posterior rami are mixed. **Posterior to anterior nuclei:** somatic sensory, visceral sensory, visceral motor, somatic motor (**S-V-V-S**).

Endo: individual axon; **peri:** fascicle; **epi:** whole nerve. Ascending sensory; descending motor. Posterior ramus serves back; anterior serves body wall/limbs and forms plexuses.

White ramus: myelinated preganglionic sympathetic fibers (T1-L2); **gray:** unmyelinated postganglionic return. "Pre/post" depends on the ganglion synapse.

Course plexus ranges: cervical C1-5; brachial C5-T1; lumbar T12-L4; sacral L4-S4. **Phrenic C3-5:** diaphragm.

Musculocutaneous: arm flexors. **Median/ulnar:** forearm/hand flexion groups. **Radial:** arm/forearm extensors. **Femoral:** anterior-thigh/hip-flexion group. **Obturator:** adductors. Superior gluteal: medius/minimus; inferior: maximus. Sciatic: highlighted hamstrings; tibial/fibular branches.

CIRCUITS + REFLEXES

Arc: receptor; sensory neuron; CNS integration; motor neuron; effector. Mono = 1 synapse/2 neurons. Poly = interneuron(s), multiple synapses.

Classify: innate/acquired; somatic/visceral; mono/poly; spinal/cranial. Innate is inborn, but brain modulation still occurs.

Stretch: spindle senses length; stretched muscle contracts. Intrafusal inside spindle; extrafusal produces force; gamma adjusts sensitivity. **Tendon reflex:** Golgi organ senses tension; inhibits muscle.

Withdrawal: flexors on, extensors off (reciprocal inhibition). **Crossed extensor:** opposite leg extensors on, flexors off. Ipsi = same; contra = opposite.

Brain: can facilitate/inhibit spinal reflexes. Reinforcement enhances. Adult plantar = toes curl; infant Babinski = fanning; adult Babinski suggests impaired descending inhibition in the chapter.

OPTIONAL HAND-DRAWN DIAGRAMS: use the blank area below. No printed images, graphs, icons or diagram templates.

Sources: supplied textbook pp. 389–464; C12/C13 lecture handouts; four class transcript files. Course threshold/ranges retained; see review-deck cautions.